

END TERM EXAMINATION

FIFTH SEMESTER (B.TECH) JANUARY-2024

Paper Code: CIC-311

Subject: Design Analysis & Algorithm

Maximum Marks: 75

Time: 3 Hours

Note: Attempt five questions including Q. no.1 which is compulsory.
Select one question from each unit.

(3×5=15)

Q1 Attempt Any five questions:

- What is an Algorithm and also discussed the characteristics of algorithm.
- Define big oh(O), Big omega(Ω) and big theta(Θ) notations
- What is the use of Warshall's algorithm and Floyd's Algorithm?
- What are NP Hard and NP Complete problems?
- Explain local search heuristic with example
- Explain optimal binary search tree Problem with example
- Apply the Quick sort to the list E, X, A, M, P, L, E

UNIT I

- Q2 a. What do you mean by complexity of an algorithm? Explain the notations used to denote the complexity of an algorithm. (7)
- b. Explain the following with example: (8)
- Selection sort.
 - Disjoint Sets.

- Q3 a. How the operations performed in Strassen's Matrix multiplication (7)
- b. Explain in detail merge sort. Illustrate the algorithm with a numeric example. Provide complete analysis of the same. (8)

UNIT II

- Q4 a. Explain Prim's algorithm for minimal spanning tree with an example. (7)
- b. Write in detail about Hamiltonian cycles. Give example to it. (8)
- Q5 a. Write backtracking algorithm for (7)
- The n-queens problem
 - Hamiltonian problem
 - The subset-Sum problem
- b. Define Greedy knapsack. Find the optimal solution of the Knapsack instance $n=7$, $M=20$, $(p_1, p_2, \dots, p_7) = (8, 5, 6, 7, 6, 12, 3)$ and $(w_1, w_2, \dots, w_7) = (2, 10, 8, 7, 6, 4, 11)$ (8)

UNIT-III

- Q6 a. Explain the Travelling salesmen problem using Branch and bound technique. (7)
- b. What is backtracking? Solve 8 queens problem with backtracking. (8)
- Q7 a. What do you mean by dynamic Programming? Explain 0/1 knapsack problem by using dynamic programming. (7)
- b. Explain the chained matrix multiplication with suitable example (8)

P.T.O.

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UNIT-IV

- Q8 a. Explain the following (7)
- NP hard Graph.
 - NP scheduling problems
- b. What are NP Hard and NP Complete problems? Explain Cook's theorem (8)
- Q9 a. Explain non-deterministic algorithm with an example. (7)
- b. What is state space tree? What are the different ways of searching an answer node in an state space tree explain with example. (8)
